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[> # Аутентификация
[> with(RandomTools) :
[> p1 := nextprime(Generate(integer(range=2..10^6)));
                                p1 := 113513 (1)
[> p2 := nextprime(Generate(integer(range=2..10^6)));
                                p2 := 958213 (2)
[> rA := p1·p2; # Открытый
                                rA := 108769632269 (3)
[> a := 53;
    igcd(a, phi(rA)); # Замкнутый
                                a := 53
                                1 (4)
[> o := a &^ (phi(phi(rA)) - 1) mod (phi(rA)); # Открытый
                                o := 10261184957 (5)
[> s := 1279856 mod rA; # Боб-->Алиса
                                s := 1279856 (6)
[> g := s &^ a mod rA; # Алиса
                                g := 87747719045 (7)
[> s1 := g &^ o mod rA; # Боб
                                s1 := 1279856 (8)
[> s1 - s; # Проверка
                                0 (9)
[>

```